Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On in Microsoft Intune

Implementation Solution Guide

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Executive Summary — Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On

A huge share of your people carry **two iPhones** — a **corporate iPhone (COD)** the company buys and controls, and their **own personal iPhone (BYOD)** they also use for work. They expect the **same work account to just sign in on both**, seamlessly. Handled badly, that means duplicate policies, inconsistent security, two logins, and frustrated users.

This solution is built for exactly that **one-person-two-iPhones** scenario, under **Microsoft Intune** and **one Microsoft Entra identity**: the user signs in with the same corporate account on **both** their company iPhone and their personal iPhone, gets **single sign-on** across them, and IT applies the **right level of control to each** based on who owns it.

- **Corporate iPhone (COD)** → supervised enrollment via **Apple Business Manager + Automated Device

Enrollment (ADE)**, full MDM, device compliance, and configuration. IT owns the iPhone end to end.

- **Personal iPhone (BYOD)** → lighter-touch management: **App Protection (MAM)** that secures

corporate data inside apps without taking over the employee's phone — or optional **Account-Driven Apple User Enrollment**.

- **One identity, both iPhones, single sign-on** → the user's single Entra account works on both

iPhones; on iOS the **Microsoft Authenticator** broker plus the **iOS Enterprise SSO extension** give **silent SSO** so they sign in once per iPhone. **Ownership-based targeting** routes the correct policy to each iPhone automatically, and **Conditional Access** grants access from their compliant company iPhone *or* a protected app on their personal one.

Built the standard repeatable way — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — with exact, click-by-click Intune steps, a requirements guide that tells you how to verify *and* fix each prerequisite, five ready-to-run PowerShell scripts (Microsoft Graph), and testing/troubleshooting guides with clear pass/fail gates.

Outcome: one employee account that signs in seamlessly — with single sign-on — across both their corporate and personal iPhone, corporate data protected on each, and access governed by compliance and app protection — delivered by your own team without a consulting engagement.

1. Solution Overview — Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On

What this solution is

This solution targets a specific, very common reality: **one person carries two iPhones** — a **personal iPhone (BYOD)** they own and a **corporate iPhone (COD)** the company owns — and needs the **same single work account to sign in seamlessly on both**, with the right level of management applied to each. Think of the executive who carries a company iPhone for work and their own iPhone for life, the clinician or field tech issued a corporate handset who still uses a personal one, or the hybrid employee who answers

Teams on either phone — all under one identity.

It delivers that in Microsoft Intune under **one Microsoft Entra identity**, across the two iOS ownership models:

1. **Corporate iPhone (COD)** — the iPhone the organization buys and owns. Enrolled **supervised via

Apple Business Manager + Automated Device Enrollment (ADE)**, fully managed in Intune (MDM); it receives configuration and compliance policies and the iOS Enterprise SSO extension.

1. **Personal iPhone (BYOD)** — the iPhone the employee owns. The corporate **identity and data** are

protected without taking over the personal phone, primarily through **App Protection Policies (MAM)**, with optional lightweight **Account-Driven Apple User Enrollment**.

The defining design goal: **the user signs in with one Entra account on both iPhones and gets a single sign-on experience** — Intune applies controls based on **device ownership**, not on a second account, so the personal iPhone is protected (data-only) while the corporate iPhone is fully managed, simultaneously, for the same person.

How it works

1. **One identity, two iPhones.** The user has a single Microsoft Entra account. That one account

enrolls their corporate iPhone **and** signs into the managed work apps on their personal iPhone — there is no separate "BYOD account," and both iPhones register to the same identity.

1. **Single sign-on across both.** On iOS the **Microsoft Authenticator** app is the identity broker;

on the corporate iPhone the **iOS Enterprise SSO extension** (deployed by Intune) extends that SSO to Safari and other apps. The user signs in once per iPhone, then moves between Microsoft 365 and SSO-enabled apps without re-entering credentials.

1. **Ownership is tagged.** Each iPhone is marked **Corporate** (ADE-enrolled) or **Personal**

(App Protection / User Enrollment defaults).

1. **Ownership-based targeting.** Policies are assigned to the **user**, then scoped with **iOS device

filters** on `deviceOwnership` so the user's two iPhones each get the correct set:

• the user's **Corporate** iPhone receives full **device compliance** + **configuration** policies

and the **Enterprise SSO extension**;

• the user's **Personal** iPhone receives **App Protection (MAM)** policies that encrypt corporate

data, enforce a PIN, and block copy/paste or "Save As" to personal locations.

1. **Conditional Access ties it together.** A single Conditional Access policy for the user grants

access when the request comes from their **compliant corporate iPhone** *or* from an **approved app with an app-protection policy** on their personal iPhone — so the one account works securely from **either** of the user's two iPhones.

Single sign-on across the user's two iPhones

Because both iPhones register to the **same Entra identity**, the user gets a unified sign-in:

- **Corporate iPhone:** supervised and Entra-registered; the **iOS Enterprise SSO extension** (the Microsoft Entra ID plug-in, brokered by Authenticator) gives **silent SSO** to Microsoft 365, Entra apps, and Safari — no repeated passwords.
- **Personal iPhone:** the **Microsoft Authenticator** broker (registered during app-protection onboarding) brokers Entra auth, so Outlook/Edge/Teams and other approved apps get silent SSO with the same account.
- **Consistent posture either way:** Conditional Access evaluates each iPhone on its own merits (compliant COD *or* MAM-protected BYOD), so the single account is honored from both without the user ever managing two logins.

Management models (choose per ownership)

- **Corporate iPhone — full MDM (supervised ADE):** complete device management, configuration profiles, compliance, and zero-touch provisioning via Apple Business Manager + Automated Device Enrollment.
- **Personal iPhone — App Protection / MAM (recommended):** secures corporate data *inside* managed apps with no full device enrollment; least intrusive for employees, strong data-leak protection.
- **Personal iPhone — Account-Driven User Enrollment (optional):** when a personal iPhone must receive a few device settings while keeping personal data separate; still tagged Personal and scoped accordingly.

This solution standardizes on **supervised full MDM for the corporate iPhone** and **App Protection for the personal iPhone**, with Account-Driven User Enrollment called out as an option.

Benefits

- **One account on both of the user's iPhones** — no duplicate identities; the same login on the personal and the corporate iPhone.
- **Seamless single sign-on** — sign in once per iPhone, then move between Microsoft 365 apps (and between the two phones) without re-entering credentials.
- **Right-sized control** — full management on the corporate iPhone; data-only protection on the personal one — at the same time, for the same person.
- **Lower friction BYOD** — employees keep their personal iPhone private; corporate data stays contained in managed apps.
- **Consistent security** — compliance for the corporate iPhone, app protection for the personal one, unified by one Conditional Access policy that accepts either iPhone.
- **Scales cleanly** — ownership-based targeting means you assign once to the user and the right policy follows each of their iPhones.

Limitations (be honest with buyers)

• **App Protection (MAM)** secures data **inside supported apps** (Microsoft 365 and SDK-enabled apps); it does not manage the whole personal iPhone or unmanaged apps.

• BYOD users must use the **managed/approved apps** (e.g. Outlook, Edge, Teams) to get corporate data; the built-in personal Mail/Safari apps are blocked from corporate data by design.

• Supervised corporate-iPhone enrollment requires the device to be in **Apple Business Manager** (ADE); iPhones not in ABM can still enroll, but not supervised/zero-touch.

• The iOS **Enterprise SSO extension** requires **Microsoft Authenticator** present on the device, and silent SSO applies to Entra-integrated apps.

• A user signing in on a brand-new iPhone enrolls/registers that device the first time.

The standard deployment model

This solution uses **one** model so the process is repeatable across our catalog:

> ***One Microsoft Entra identity per user, signed in across that user's two iPhones with single sign-on; the corporate iPhone under supervised full MDM (ABM + ADE) and the personal iPhone under App Protection (MAM); ownership-based targeting routes the right policy to each; a single Conditional Access policy governs access from both.***

Every step follows the repeatable pattern: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

2. Requirements — Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On

Confirm **every** requirement below before you begin. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.** Category-wide prerequisites (tenant, Intune licensing, admin roles, pilot group, supported OS, network) are covered once in the **Global Requirements — Intune Solutions** document; the items here are specific to running a corporate iPhone (COD) and a personal iPhone (BYOD) under one identity.

R1. A single Microsoft Entra identity per user

Description: Each user has exactly one Entra ID account used for both device types. **Why:** The whole design depends on one account signing in on both the corporate and the personal device; two accounts would split policy and defeat the goal. **Verify:** Entra → **Users**; the pilot user exists once with a licensed account. **Implement:** Consolidate duplicate accounts; ensure the user signs in with the same UPN everywhere.

R2. Intune licensing that includes App Protection (MAM)

Description: A license plan that covers both device management (MDM) and app management (MAM). **Why:** COD uses MDM; BYOD uses App Protection (MAM). Both must be licensed for the same user. **Verify:** Entra → user → **Licenses**; an Intune-bearing plan (M365 E3/E5, Business Premium, or Intune Plan 1) is assigned. **Implement:** Assign the appropriate Microsoft 365/Intune license to all pilot users.

R3. Microsoft Entra ID P1 (for Conditional Access)

Description: Entra ID P1 (or higher) for Conditional Access. **Why:** A single Conditional Access policy is what lets one account access work from a compliant COD *or* an app-protected BYOD device; CA requires P1. **Verify:** Entra → **Conditional Access** opens and policies can be created (P1 present). **Implement:** Assign Entra ID P1 (included in M365 E3/E5/Business Premium) to pilot users.

R4. Enrollment restrictions configured for both ownership types

Description: Device platform/enrollment restrictions that **allow personal (BYOD)** enrollment for the BYOD path and treat the correct platforms as **corporate** for COD. **Why:** A restriction that blocks personal enrollment, or that auto-marks all devices corporate, breaks the ownership distinction this solution relies on. **Verify:** Intune → **Devices** → **Enrollment** → **Enrollment device platform restrictions**; the pilot restriction shows the intended **Personal = Allow/Block** per platform. **Implement:** Edit the default or a pilot-targeted restriction so BYOD platforms allow personal, and keep COD platforms enrolling as corporate (see R5).

R5. A way to mark the corporate iPhone Corporate (ABM + ADE)

Description: The corporate iPhone enrolled through **Apple Business Manager + Automated Device Enrollment (ADE)** so it is tagged **Corporate** and supervised. **Why:** Ownership = Corporate is the signal that routes full MDM policy (vs. App Protection) to the iPhone; ADE also makes the enrollment supervised and zero-touch. **Verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens** → **(token)** → **Devices** shows the iPhone serials; a device's **Ownership** field reads **Corporate**. **Implement:** Add the iPhone serials to ABM and assign them to the Intune MDM server (Implementation §§3, 7), or for non-ABM iPhones add **Corporate device identifiers** (serials/IMEIs).

R6. Microsoft Authenticator and Company Portal apps

Description: Microsoft Authenticator (the iOS identity broker) and the Intune **Company Portal** available on the iPhones. **Why:** Authenticator brokers Entra auth and powers the iOS Enterprise SSO extension; Company Portal surfaces app-protection-enabled apps and supports User Enrollment for BYOD. **Verify:** Intune → **Apps** → **iOS/iPadOS**; Authenticator and Company Portal are added and assigned to the pilot group. **Implement:** Add both apps for iOS/iPadOS and assign them to the pilot group.

R7. App Protection (MAM) capability and approved apps

Description: App Protection Policies plus the approved managed apps (Outlook, Edge, Teams, Office). **Why:** BYOD data protection lives in App Protection Policies applied to managed apps; without the apps, BYOD users can't safely reach corporate data. **Verify:** Intune → **Apps** → **App protection policies** can be created; the approved apps are deployed or available. **Implement:** Create App Protection Policies (Implementation §4) and make the approved apps available to BYOD users via Company Portal / store.

R8. Device compliance policy for corporate devices

Description: A compliance policy targeting corporate-owned devices. **Why:** Conditional Access for COD grants access only to **compliant** devices; the policy defines compliant. **Verify:** Intune → **Devices** → **Compliance**; a policy exists and is assigned (scoped to Corporate). **Implement:** Create a compliance policy (OS minimums, encryption, etc.) and target it with the Corporate device filter (Implementation §5).

R9. Device-ownership filters

Description: Intune **device filters** on `deviceOwnership` (Corporate / Personal). **Why:** Filters are how one assignment to a user group applies the *right* policy to the *right* device type — the mechanism that makes "one account, both devices" work cleanly. **Verify:** Intune → **Tenant administration** → **Filters**; the `iPhone – Corporate` and `iPhone – Personal` filters exist. **Implement:** Create them in Implementation §5, or run the included `IOSDD-Set-OwnershipBasedTargeting.ps1`.

R10. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, app protection check-ins, and authentication depend on these. **Verify:** From an iPhone, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

R11. Apple MDM Push (APNs) certificate

Description: A valid **Apple MDM Push certificate (APNs)** in Intune. **Why:** Every iOS enrollment — supervised corporate ADE *and* personal User Enrollment — and ongoing MDM communication requires APNs. **Verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate** shows **Active** with a future expiration. **Implement:** Create/renew it (Implementation §2) using a **company Apple ID** you reuse each year.

R12. Apple Business Manager + ADE token

Description: An **Apple Business Manager** account with the corporate iPhones added, and an **Automated Device Enrollment token** connected to Intune. **Why:** Supervised, zero-touch corporate-iPhone enrollment depends on ABM holding the device serials and a valid ADE token mapping them to the Intune MDM server. **Verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens** shows a token with a future expiration and the iPhone serials under **Devices**. **Implement:** Connect ABM and upload the ADE token (Implementation §3); add device serials in ABM and assign them to the Intune MDM server.

R13. iOS Enterprise SSO extension capability

Description: The ability to deploy the **iOS Enterprise SSO extension** (Microsoft Entra ID plug-in) via a Device features configuration profile. **Why:** It delivers silent SSO to Safari and Entra apps on the corporate iPhone, brokered by Authenticator — the SSO half of "one account, both iPhones." **Verify:** Intune → **Devices** → **Configuration** → **Create** → **iOS/iPadOS** → **Templates** → **Device features** exposes **Single sign-on app extension**. **Implement:** Create the profile in Implementation §12 and ensure Authenticator (R6) is deployed.

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level on the marketplace so individual solution listings stay small. Confirm these are in place before deploying *any* Intune solution.

Each entry: **Description · Why required · How to verify · How to implement if missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, enrollment, and security policies. **Verify:** `https://intune.microsoft.com` loads and **Tenant administration** → **Tenant status** shows an active subscription. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies, enrollment connectors, and configuration profiles requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator** (or Global Administrator for connector setup). **Implement:** Assign **Intune Administrator** to the operator; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted devices are enrolled and manageable in Intune. **Why:** Policies only reach enrolled (or app-managed) devices, regardless of platform. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Configure the appropriate enrollment method for the platform and ownership model, then enroll the device.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted to a small group before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists with intended members. **Implement:** Entra → **Groups** → **New group** → type **Security** → add pilot members.

G6. Supported, current operating system

Description: A supported OS baseline with current updates for the targeted platform. **Why:** Modern Intune features and security settings assume a patched, vendor-supported OS. **Verify:** Check the device's OS version or Intune → device → **Hardware**. **Implement:** Deploy update policies or upgrade to a supported release.

G7. Network access to required cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation — Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On

Follow these steps **in order**. The *why* lives in the Solution Overview; the steps below are short and exact so you can **click to implement**. The whole design assumes **one Entra identity per user** who carries **two iPhones** — a **corporate iPhone (COD)** and a **personal iPhone (BYOD)** — so you assign policies to the **user (group)** and let **ownership-based filters** route the right policy to each iPhone. Pilot first: **Assess → Configure → Pilot → Validate → Expand.**

Consoles used:

- **Microsoft Intune admin center** — `https://intune.microsoft.com`
- **Microsoft Entra admin center** — `https://entra.microsoft.com`
- **Apple Business Manager (ABM)** — `https://business.apple.com`

> ***Scope note (iOS iPhones):*** *the* ***corporate iPhone*** *is enrolled supervised via* ***Apple Business Manager + Automated Device Enrollment (ADE)*** *for full MDM. The* ***personal iPhone*** *is protected by* ***App Protection Policies (MAM)*** *— with* ***Account-Driven Apple User Enrollment*** *called out as an optional lightweight MDM path.* ***Single sign-on*** *across both iPhones is delivered by the* ***Microsoft Authenticator*** *broker plus the* ***iOS Enterprise SSO extension****, all under one Entra identity.*

Phase 1 — Identity, Apple connection & enrollment baseline

Step 1. Confirm one identity and licensing per pilot user

1. Entra → **Users** → confirm each pilot user is a **single** account with an Intune + Entra ID P1

license (Requirements R1–R3).

1. Entra → **Groups** → **New group** → **Security** → name it `PILOT – iPhone COD-BYOD Users` → add

the pilot users. *(You assign every policy in this solution to this **user** group.)*

Step 2. Create the Apple MDM Push (APNs) certificate

1. In Intune, go to **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate**.
2. Check **I agree**, then **Download** the CSR file.
3. Click **Create your MDM push certificate** → `https://identity.apple.com`; sign in with a

company Apple ID (record which Apple ID — you must reuse it every year).

1. **Create a Certificate**, upload the CSR, **Download** the `.pem`, then back in Intune enter that **Apple ID**, **Browse** to the `.pem`, and **Upload**.

1. Confirm the status shows **Active** with an expiration ~1 year out. *(Required for both corporate-iPhone MDM and personal-iPhone User Enrollment.)*

Step 3. Connect Apple Business Manager for corporate iPhones (ADE token)

1. In Intune, go to **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens** → **Add** → check **I agree** → **Download** the Intune **public key** (`.pem`).

1. In **ABM** → **Preferences** → **MDM Servers** → **Add**; name it `Intune – iPhone COD`; **Upload** the Intune public key; **Save**; **Download Token** (`.p7m`).

1. Back in Intune, enter the **Apple ID** used for ABM, **Browse** to the `.p7m`, **Upload**. The token appears under **Enrollment program tokens** with a future expiration.

Step 4. Set enrollment restrictions for iPhone ownership types

1. **Devices** → **Enrollment** → **Enrollment device platform restrictions.**

2. Open the default (or create a pilot-scoped restriction **named `Enrollment Restriction – iPhone COD-BYOD – Pilot`**).

1. For **iOS/iPadOS**, set **Personal devices = Allow** *(so the user's personal iPhone may enroll for User Enrollment / register for App Protection)*.

1. Corporate iPhones are forced through ADE in the next phase, so they always tag **Corporate**.

Phase 2 — Ownership-based targeting (the core of "one account, both iPhones")

Step 5. Create device-ownership filters

1. **Tenant administration** → **Filters** → **Create.**

2. Filter 1: **name it** `iPhone – Corporate` — platform **iOS/iPadOS**; rule `(device.deviceOwnership -eq "Corporate")`.

1. Filter 2: **name it** `iPhone – Personal` — platform **iOS/iPadOS**; rule `(device.deviceOwnership -eq "Personal")`.

1. **Save** both.

Prefer automation? Run `IOSDD-Set-OwnershipBasedTargeting.ps1` to create these filters (and optionally assign the policies below) via Microsoft Graph.

Phase 3 — Corporate iPhone (COD): supervised ADE + full MDM

Step 6. Create the iOS enrollment program profile (supervised)

1. In Intune, go to **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens**, **open** your ADE token, then **Profiles** → **Create profile** → **iOS/iPadOS**; enter a name and description — **name it** `iPhone ADE – COD – Pilot`.

1. **User affinity** = **Enroll with user affinity** (required so the corporate iPhone ties to the user and can do SSO). Choose **Setup Assistant with modern authentication**.

1. **Management options:** enable **Supervised**; set **Locked enrollment = Yes** (prevents users removing management).

1. **Setup Assistant:** hide unnecessary panes (Restore, Apple ID, Siri, Privacy as policy dictates) for a clean first-run. **Review + Create.**

Step 7. Assign devices and the profile (in ABM + Intune)

1. In **ABM** → **Devices**, select the corporate iPhone serial numbers → **Edit MDM Server** → choose `Intune – iPhone COD` → **Confirm**.

1. In Intune, on the token → **Devices**, click **Sync** (Apple rate-limits; allow a few minutes), then select the serials → **Assign profile** → `iPhone ADE – COD – Pilot`.

Step 8. Create a compliance policy for corporate iPhones

1. **Devices** → **Compliance** → **Create policy** → **iOS/iPadOS**; **name it `iPhone – Compliance – COD – Pilot`**.

1. Set baseline rules (min iOS version, **require passcode**, jailbreak detection, etc.).

2. **Assignments:** assign to `PILOT – iPhone COD-BYOD Users`; under **Filter**, **Include** the `iPhone – Corporate` filter → **Create**.

Step 9. Create a configuration profile for corporate iPhones

1. **Devices** → **Configuration** → **Create** → **New policy** → **iOS/iPadOS**; **name it `iPhone – Configuration – COD – Pilot`** (settings such as passcode, restrictions, Wi-Fi/VPN as needed).

1. **Assignments:** assign to `PILOT – iPhone COD-BYOD Users`, **Include** the **`iPhone – Corporate`** filter.

Phase 4 — Personal iPhone (BYOD): App Protection (MAM)

Step 10. Create the iOS App Protection Policy

1. **Apps** → **App protection policies** → **Create policy** → **iOS/iPadOS**; **name it `iPhone – App Protection – BYOD – Pilot`**.

1. **Apps:** select the approved managed apps (Outlook, Edge, Teams, Office, OneDrive, **Microsoft Authenticator**).
1. **Data protection:** block **backup**, restrict **cut/copy/paste** to managed apps, block **Save As** to personal locations, keep corporate data in managed apps.
1. **Access requirements:** require **PIN** / org credentials; set offline timeouts.
2. **Assignments:** assign to `PILOT – iPhone COD-BYOD Users`. *(App Protection is identity-based — it applies wherever the user signs into the managed app, so it targets the personal iPhone with no device enrollment.)*

Step 11. Make the approved apps available to BYOD users

1. **Apps** → **iOS/iPadOS** → **Add** Outlook, Edge, Teams, Office, OneDrive, **Company Portal**, and **Microsoft Authenticator**.
1. Assign them **Available** to `PILOT – iPhone COD-BYOD Users` so users install them from the App Store / Company Portal.

> ***Optional lightweight MDM for the personal iPhone:*** *if a personal iPhone must receive a few device settings, use* ***Account-Driven Apple User Enrollment*** *(Devices → Enrollment → Apple → enable* ***Account driven user enrollment****,* ***name the profile*** `iPhone – User Enrollment – BYOD – Pilot`*). It still tags* ***Personal*** *and is scoped by the* `iPhone – Personal` *filter, keeping personal data separate.*

Phase 5 — Single sign-on across the user's two iPhones

The goal: the user authenticates **once per iPhone**, then gets **silent SSO** to Microsoft 365 and Entra apps on **both** iPhones with the same account. On iOS the **Microsoft Authenticator** app is the identity broker, and the **iOS Enterprise SSO extension** extends that SSO to Safari and other apps.

Step 12. Deploy the iOS Enterprise SSO extension to the corporate iPhone

1. **Devices** → **Configuration** → **Create** → **New policy** → **iOS/iPadOS** → **Templates** → **Device features**;

name it `iPhone – Enterprise SSO – COD – Pilot`.

1. Under **Single sign-on app extension**, set **SSO app extension type = Microsoft Entra ID** (Microsoft's Enterprise SSO plug-in, brokered by Authenticator).
1. Leave the Microsoft-managed defaults; under **App bundle IDs**, add Safari and any line-of-business apps that should get silent SSO.
1. **Assignments:** assign to `PILOT – iPhone COD-BYOD Users`, **Include** the **`iPhone – Corporate` **filter** → Create**.

Step 13. Establish the Authenticator broker on the personal iPhone

1. On the personal iPhone, the user installs **Microsoft Authenticator** (it is the **identity

broker** that App Protection onboarding registers).

1. The first sign-in to a managed app (Outlook/Edge/Teams) registers the account through the broker;

afterward, **other approved apps get silent SSO** with the same account.

1. Do **not** require enrollment for SSO — the broker provides SSO while the personal iPhone stays

BYOD (App Protection only).

Step 14. Confirm the same identity owns both iPhones

1. Entra → **Devices** → filter by the pilot user: confirm the **same UPN** has the **corporate

iPhone **(Entra-registered, supervised)** and **a** registered personal iPhone** under one identity.

1. The user should move between their two iPhones and into M365 apps **without re-entering

credentials** beyond one sign-in per iPhone.

Phase 6 — Unify access with Conditional Access

Step 15. One Conditional Access policy for both iPhones

1. Entra → **Protection** → **Conditional Access** → **Create new policy**; **name it `CA – iPhone COD-BYOD

Access`**.

1. **Users:** `PILOT – iPhone COD-BYOD Users`.
2. **Target resources:** Office 365 (and other corporate apps).
3. **Conditions** → **Device platforms: iOS**.
4. **Grant:** require **one of the following** — **Require device to be marked as compliant**

(satisfied by the corporate iPhone) **and/or Require app protection policy** *(satisfied by the personal iPhone's managed app)*; set to **Require one of the selected controls**.

1. **Enable policy: Report-only** for the pilot → **Create**.

This single policy lets the **same account** reach corporate data from a **compliant corporate iPhone** or an **app-protected personal iPhone**.

Phase 7 — Pilot, validate, expand

Step 16. Pilot with one user on two iPhones

1. **Erase** the pilot **corporate** iPhone (**Settings → General → Transfer or Reset → Erase All

Content and Settings**) so it returns to Setup Assistant; enroll via ADE — it tags** Corporate** and receives compliance/config + the Enterprise SSO extension.

1. On the **personal** iPhone, install **Microsoft Authenticator**, then sign into **Outlook/Edge**

with the **same account** → the App Protection policy applies (PIN prompt, managed-app containment).

1. Confirm the **same UPN** now owns both a Corporate and a Personal iPhone in Intune.

Step 17. Validate

1. Run `IOSDD-Discovery.ps1` (tenant readiness: APNs, ADE token, filters, App Protection, CA) and `IOSDD-Validation.ps1 -User <upn>`: confirm the user has **one Corporate iPhone** (compliant, supervised) and **one Personal iPhone** (app protection applied), under a single identity.

1. Confirm the `iPhone – Corporate` filter applied device policy and the `iPhone – Personal` filter did not (and vice-versa for app protection).

1. **Verify single sign-on:** on the corporate iPhone open a Microsoft 365 web app in **Safari** and confirm **silent SSO** (no password prompt — the Enterprise SSO extension at work); on the personal iPhone, after the first managed-app sign-in, a second approved app (e.g. Teams) opens **without** a fresh password — silent SSO on both iPhones for the one account.

1. Review Conditional Access **report-only** results for the pilot user.

Step 18. Expand

1. When the pilot passes, widen the assignments from the pilot group to broader user groups (or make `iPhone ADE – COD – Pilot` the default profile for the ADE token).

1. Move `CA – iPhone COD-BYOD Access` from **report-only** to **On**.

2. Roll out in waves, validating each wave with the scripts before widening.

5. Testing & Validation — Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On

Validate against a **pilot user with two iPhones** (one corporate, one personal) before expanding. Each test has an explicit **pass/fail gate**. Use the included PowerShell scripts (Microsoft Graph) for tenant-side checks and the on-device steps for the user experience.

Test 1 — Tenant readiness

Procedure: Run `IOSDD-Discovery.ps1`. **Pass:** The APNs certificate is **Active**, the ADE token is present, enrollment restrictions allow the intended Personal/Corporate split, an App Protection policy exists, a compliance policy exists, the `iPhone – Corporate`/`iPhone – Personal` **device filters** exist, Authenticator + Company Portal are assigned, and a Conditional Access policy is present. **Fail:** Any missing → resolve before piloting.

Test 2 — Single identity, two iPhones

Procedure: Run `IOSDD-Validation.ps1 -User <upn>` for the pilot user. **Pass:** The **same UPN** owns at least one **Corporate** iPhone and one **Personal** iPhone. **Fail:** Devices split across different accounts → consolidate identity (Requirements R1).

Test 3 — Corporate iPhone gets full MDM

Procedure: Inspect the pilot **corporate** iPhone in Intune. **Pass:** Ownership = **Corporate**, **Supervised = Yes**; the `iPhone – Compliance – COD – Pilot` and `iPhone – Configuration – COD – Pilot` policies (Corporate filter) are **applied**; device reports **Compliant**. **Fail:** Policies not applied → confirm the iPhone is tagged Corporate via ADE (R5) and the `iPhone – Corporate` filter is set on the assignment (Implementation §§5, 8–9).

Test 4 — Personal iPhone gets App Protection only

Procedure: On the pilot **personal** iPhone, sign into Outlook/Edge with the pilot account. **Pass:** The `iPhone – App Protection – BYOD – Pilot` policy applies — **PIN prompt**, corporate data confined to managed apps, **Save As**/paste to personal locations blocked. The iPhone is **not** fully enrolled (or, if User-Enrolled, is tagged **Personal** and does **not** receive Corporate device policies). **Fail:** No PIN/containment → confirm the app is approved/managed and the App Protection policy is assigned to the user (Implementation §10).

Test 5 — Ownership filters route correctly

Procedure: Compare policy application across the two iPhones. **Pass:** The **Corporate** iPhone receives device compliance/config and **not** the BYOD-only items; the **Personal** iPhone receives app protection and **not** corporate device policies. **Fail:** Cross-application → re-check the `deviceOwnership` filter rules and the Include/Exclude on each assignment (Implementation §5).

Test 6 — Single sign-on across both iPhones

Procedure: On the **corporate** iPhone, open a Microsoft 365 web app in **Safari**; on the **personal** iPhone, after the first managed-app sign-in, open a second approved app (e.g. Teams). **Pass:** The corporate iPhone gets **silent SSO** (no password prompt — the iOS Enterprise SSO extension at work); the personal iPhone's second app opens **without** a fresh password (the Authenticator broker at work) — one account, silent SSO on both. **Fail:** Repeated prompts → confirm Microsoft Authenticator is installed on both iPhones and the `iPhone – Enterprise SSO – COD – Pilot` profile is applied to the corporate iPhone (Implementation §§12–13).

Test 7 — One account reaches work from both iPhones

Procedure: With Conditional Access **report-only**, access a corporate web app from the corporate iPhone and a managed app from the personal iPhone, signed in as the same user. **Pass:** Both succeed — the corporate iPhone satisfies **compliant device**; the personal iPhone satisfies **app protection policy** — under the **single** CA policy. **Fail:** One path blocked → review CA grant controls ("require one of"), compliance state, and app protection assignment (Implementation §15).

Test 8 — Unmanaged path is blocked (data protection holds)

Procedure: On the personal iPhone, try to open corporate mail in the **built-in** (unmanaged) Mail app or paste corporate data into a personal app. **Pass:** Corporate data is **not** accessible in unmanaged apps; paste/Save-As to personal locations is blocked. **Fail:** Corporate data leaks to unmanaged apps → tighten App Protection data-relocation settings.

Test 9 — Compliance & enforcement

Procedure: Review compliance for the corporate iPhone and the report-only CA results for the user. **Pass:** Corporate iPhone **Compliant**; CA evaluates the user correctly from both iPhones. **Fail:** Non-compliant or CA misfires → review the compliance policy and CA conditions.

Acceptance gate

Proceed to **Expand** only when Tests 1–9 pass for a pilot user who has **both** a corporate and a personal iPhone signed in with the **same account**, single sign-on works on both (Test 6), and the data-protection checks (Test 8) hold.

6. Troubleshooting — Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On

Work top-down: confirm the **identity** (one account), then **ownership** (Corporate vs Personal), then **targeting** (filters/assignments), then **SSO**, then **access** (Conditional Access). Most issues are an iPhone tagged with the wrong ownership, a missing filter on an assignment, a BYOD user not using a managed app, or Microsoft Authenticator not installed.

A user's two iPhones show under different accounts

- **Duplicate identities.** Consolidate to one Entra account; the user must sign in with the same UPN on both iPhones (Requirements R1). Re-enroll/re-register the stray iPhone under the correct account.

A corporate iPhone is tagged Personal (or vice-versa)

- **Wrong ownership.** Intune → **Devices** → **(device)** → **Properties** → **Device ownership** → set **Corporate**/**Personal**. For the corporate iPhone, enroll it via **ABM + ADE** so it tags Corporate (and supervised) automatically (R5/R12).
- After fixing, the iPhone picks up the correct filtered policies on next sync.

Corporate policies are not applying to the corporate iPhone

- **Filter mismatch.** The compliance/config assignment must **Include** the `iPhone – Corporate` filter and target the **user** group (Implementation §§8–9). Verify the filter rule is `(device.deviceOwnership -eq "Corporate")`.
- **Device not compliant/enrolled.** Confirm the iPhone is enrolled (supervised, ADE) and checking in.

App Protection isn't applying to the personal iPhone

- **User not in scope.** App Protection assigns to the **user** group — confirm the pilot user is a member (Implementation §10).
- **App not managed/approved.** The user must use an **approved managed app** (Outlook, Edge, Teams).

The built-in Mail/Safari apps are intentionally excluded.

- **No Intune/MAM license** on the user (R2).
- App Protection applies on **next app launch / token refresh** — close and reopen the managed app.

Personal iPhone user can't access corporate mail/data

- **By design in an unmanaged app.** They must use the managed app (e.g. Outlook), not the built-in one. Deploy and direct them to the approved apps (Implementation §11).
- **Conditional Access blocking.** If CA requires app protection, accessing from a non-managed app is blocked — expected. Check Entra → **Sign-in logs** for the CA result.

Single sign-on keeps prompting for a password

- **Authenticator missing.** Microsoft Authenticator is the iOS identity broker — it must be installed on **both** iPhones (R6). On the personal iPhone, sign into one managed app first to register it.
- **Enterprise SSO extension not applied.** On the corporate iPhone, confirm the `iPhone – Enterprise SSO – COD – Pilot` profile shows **Succeeded** (Intune → Devices → (device) → Device configuration) and that Safari/the target app's bundle ID is listed (Implementation §12).
- **App not Entra-integrated.** Silent SSO applies to Entra-integrated apps; a non-Entra app will still prompt.

One account works on the corporate iPhone but not the personal one (or vice-versa)

- **CA grant too strict.** The Conditional Access policy must **require *one of*** compliant device **or** app protection — not **both** (Implementation §15). "Require all" blocks one of the two paths.
- **App protection not yet registered** on the personal iPhone — open a managed app and complete the PIN setup.

Personal iPhone received corporate device policies (over-management)

- The personal iPhone was **fully enrolled** and/or tagged **Corporate**. Confirm enrollment restrictions and ownership (R4–R5); for BYOD, prefer **App Protection without enrollment**. If Account-Driven User Enrollment is intended, ensure it's tagged **Personal** so the `iPhone – Personal` filter excludes corporate policies.

Corporate iPhone won't enroll via ADE

- **APNs/ADE token lapsed.** Renew the **Apple MDM Push certificate** (R11) and the **ADE token** (R12) — an expired token blocks all iOS enrollment.

- **Serial not assigned.** Confirm the iPhone serial is in ABM and assigned to the `Intune – iPhone COD`

MDM server, and the `iPhone ADE – COD – Pilot` profile is assigned (Implementation §§3, 7).

Conditional Access has unexpected effects

- Keep the pilot policy in **report-only** and read Entra → **Sign-in logs** → **Conditional Access** to

see exactly which control was/wasn't satisfied before turning it **On**.

Useful checkpoints

- **Intune:** Devices → (device) → **Properties** (ownership), **Compliance**, **Configuration**;

Apps → **App protection policies** → **(policy)** → **Monitor** for per-user status; Tenant administration → **Filters**; Devices → Enrollment → Apple for APNs/ADE token status.

- **Entra: Sign-in logs** → **Conditional Access** tab for grant evaluation.
- **Scripts:** re-run `IOSDD-Validation.ps1 -User <upn>` after each change to confirm the gate.

7. Best Practices — Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On

Field-proven guidance to keep one identity working cleanly across both device types.

Identity & assignment model

- **Assign policies to user groups, scope with device filters.** This is the heart of the design:

one assignment to the user, the right policy follows the device's ownership. Avoid mixing per-device group assignments that fight the filters.

- **Keep one Entra identity per user.** Never create a separate "BYOD account" — it splits policy and

breaks Conditional Access logic.

- **Name groups and filters by intent** (e.g. *PILOT – iPhone COD-BYOD Users*, *iPhone – Corporate*,

iPhone – Personal) so assignments are self-documenting.

Ownership hygiene

- **Make ownership deterministic.** Enroll the corporate iPhone through **ABM + ADE** so it tags

Corporate automatically; let the personal iPhone default to **Personal**.

- **Audit ownership regularly** with `IOSDD-DeviceReport.ps1` — a device tagged the wrong way gets

the wrong policy set.

BYOD data protection

• **Prefer App Protection (MAM) without enrollment for BYOD.** It protects corporate data without managing the employee's personal device — better adoption, less privacy friction, strong data-leak controls.

• **Lock the data boundary:** block backup, restrict cut/copy/paste and "Save As" to managed apps, require a PIN, and enforce encryption within managed apps.

• **Only enroll the personal iPhone (Account-Driven User Enrollment) when you truly need device settings **— and keep it tagged** Personal** so it never receives corporate device policies.

Corporate iPhone management

• **Apply a real compliance baseline** (min iOS version, passcode, jailbreak detection) and gate access on it via Conditional Access.

• **Use Apple Business Manager + ADE** for hands-off, supervised corporate-iPhone provisioning, and **renew the APNs and ADE tokens** before they expire (a lapse blocks all iOS enrollment).

Conditional Access

• **One policy, "require one of."** Grant access when the request is from a **compliant** device **or** an **app-protection-protected** app — never require both, or you lock out one device type.

• **Always start report-only.** Read the sign-in logs for the pilot user before enforcing.

• **Add MFA** as part of the same access story for a complete Zero Trust posture.

Rollout discipline

• **Pilot one user with two real devices** (a corporate and a personal one) — it's the only way to prove the single-identity experience end to end.

• **Expand in waves**, validating each with `IOSDD-Validation.ps1` before widening.

• **Communicate the BYOD deal to employees:** what IT can and cannot see/do on a personal device under App Protection (data containment, *not* device surveillance) — clarity drives adoption.

Operations & documentation

• **Keep a runbook** of the pilot/production groups, the two ownership filters, the compliance and app-protection policies, and the Conditional Access policy — the moving parts that define the model.

• **Report monthly** with `IOSDD-DeviceReport.ps1`: device counts by ownership, users who have both a COD and a BYOD device, compliance, and app-protection coverage.

8. Deliverables Summary — Dual-Device iPhone (Personal + Corporate iOS) Management with Single Sign-On

Everything included in this solution package, and what each piece is for.

Documentation (this guide)

- **Executive Summary** — the business case and outcome in one page.
- **Solution Overview** — the two ownership models, the single-identity design, management models, benefits, and honest limitations.
- **Requirements Guide** — every prerequisite (one identity, MDM+MAM licensing, Entra ID P1, enrollment restrictions, corporate tagging, Company Portal, App Protection, compliance, ownership filters, network) with how to **verify** and how to **implement if missing**, plus the category-level Global Requirements.
- **Implementation Guide** — exact, ordered, click-by-click steps across seven phases: identity, Apple connection & enrollment baseline, ownership-based targeting, corporate-iPhone supervised ADE full-MDM, personal-iPhone App Protection, single sign-on across both iPhones, unifying access with Conditional Access, then pilot/validate/expand.
- **Testing & Validation** — eight tests with explicit pass/fail gates and an acceptance gate, proving one account works on both a corporate and a personal device.
- **Troubleshooting** — top-down diagnosis for identity, ownership, targeting, and access failures.
- **Best Practices** — the assign-to-user/scope-by-filter model, ownership hygiene, BYOD data protection, COD management, Conditional Access, and rollout discipline.

PowerShell scripts (Microsoft Graph) — `IOSDD-Scripts.zip`

All scripts are tenant-side (run from an admin workstation with the Microsoft Graph PowerShell SDK); each has a header explaining **Purpose · How to run · Expected results**.

1. `IOSDD-Discovery.ps1` — tenant readiness: enrollment restrictions (personal allowed?), compliance policies, App Protection policies, Corporate/Personal device filters, Company Portal, and Conditional Access presence.
1. `IOSDD-Validation.ps1` — for a given user, confirms a single identity owns both a **Corporate** device (compliant) and a **Personal** device (app protection applied), with a PASS/FAIL summary.
1. `IOSDD-DeviceReport.ps1` — tenant-wide CSV: devices by **ownership**, platform, enrollment type, compliance, and primary user — plus a list of users who have **both** a COD and a BYOD device.
1. `IOSDD-Remediation.ps1` — safe remediation: set device **ownership**, re-assign policies to the target group, and flag enrollment-restriction/filter misconfigurations. Read-only by default; changes require an explicit confirmation switch.

1. `IOSDD-Set-OwnershipBasedTargeting.ps1` — creates the `iPhone – Corporate` and **`iPhone –

Personal** deviceOwnership` filters (and optionally assigns the compliance/app-protection policies with those filters) — the mechanism that makes one account work cleanly across both iPhones.

Hero image

- `IOSDD-hero.png` — the branded product image (1200×1200) used on the marketplace listing.

How it's delivered

A branded **Word** document and **PDF** (cover, table of contents, page numbers, and a scripts appendix), plus all five PowerShell scripts in a single **ZIP** — yours to keep and reuse across **unlimited devices**.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

IOSDD-DeviceReport.ps1

Purpose: Exports every managed device with ownership (Corporate/Personal), platform, enrollment type, compliance, and primary user — then lists users who own BOTH a corporate and a personal device (the population this solution is designed for).

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\IOSDD-DeviceReport.ps1 .\IOSDD-DeviceReport.ps1 -CsvPath C:\Temp\cod-byod.csv

IOSDD-Discovery.ps1

Purpose: Confirms the building blocks of the one-identity / two-ownership model exist: enrollment restrictions, a compliance policy (for COD), App Protection policies (for BYOD), the Corporate/Personal device filters, Company Portal, and a Conditional Access policy.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\IOSDD-Discovery.ps1 Reads only — makes no changes.

IOSDD-Remediation.ps1

Purpose: The common fixes that unblock the COD/BYOD model: correct a device's ownership (Corporate/Personal) so the right filtered policies apply, and audit for devices whose ownership is Unknown. Read-only by default.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once # Audit only (no changes): .\IOSDD-Remediation.ps1 # Set a device's ownership (explicit opt-in): .\IOSDD-Remediation.ps1 -Apply -DeviceName "Jane-Surface" -Ownership company

IOSDD-Set-OwnershipBasedTargeting.ps1

Purpose: Creates the two Intune assignment filters on `deviceOwnership` that make "one account, both device types" work: a Corporate filter (routes full-MDM policy to company devices) and a Personal filter (keeps corporate device policy off BYOD). Idempotent — skips a filter that already exists.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\Set-OwnershipBasedTargeting.ps1 -WhatIf # preview .\Set-OwnershipBasedTargeting.ps1 # create

IOSDD-Validation.ps1

Purpose: For a given user (UPN), lists their Intune devices and validates the acceptance gate: the same account owns at least one Corporate device (compliant) and at least one Personal device, and an App Protection policy targets the user. Produces a clear PASS/FAIL verdict.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\IOSDD-Validation.ps1 -User jane@contoso.com